1. The x position of a train over time is given by: $x(t) = 3t^2 - 2t + 1$. The goal of this exercise is to determine the instantaneous velocity at 2 seconds. Recall that:

$$v(t) = \lim_{h \to 0} \frac{x(t+h) - x(t)}{h}$$

At t=2 this simplifies to

$$v(2) = \lim_{h \to 0} \frac{x(2+h) - x(2)}{h}$$

A. Expand out x(2+h) for the function above

B. Compute x(2)

C. Determine v(2)